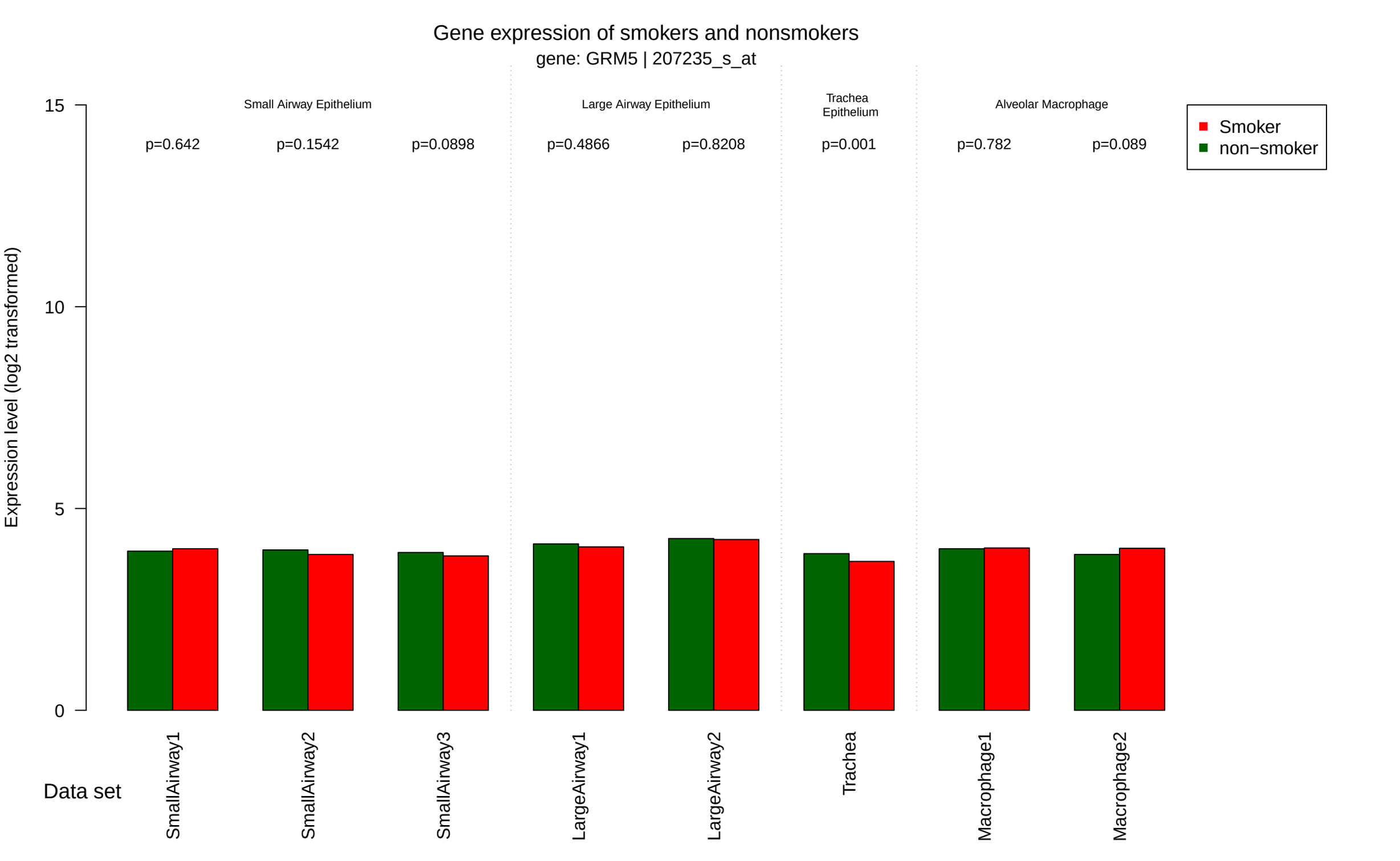
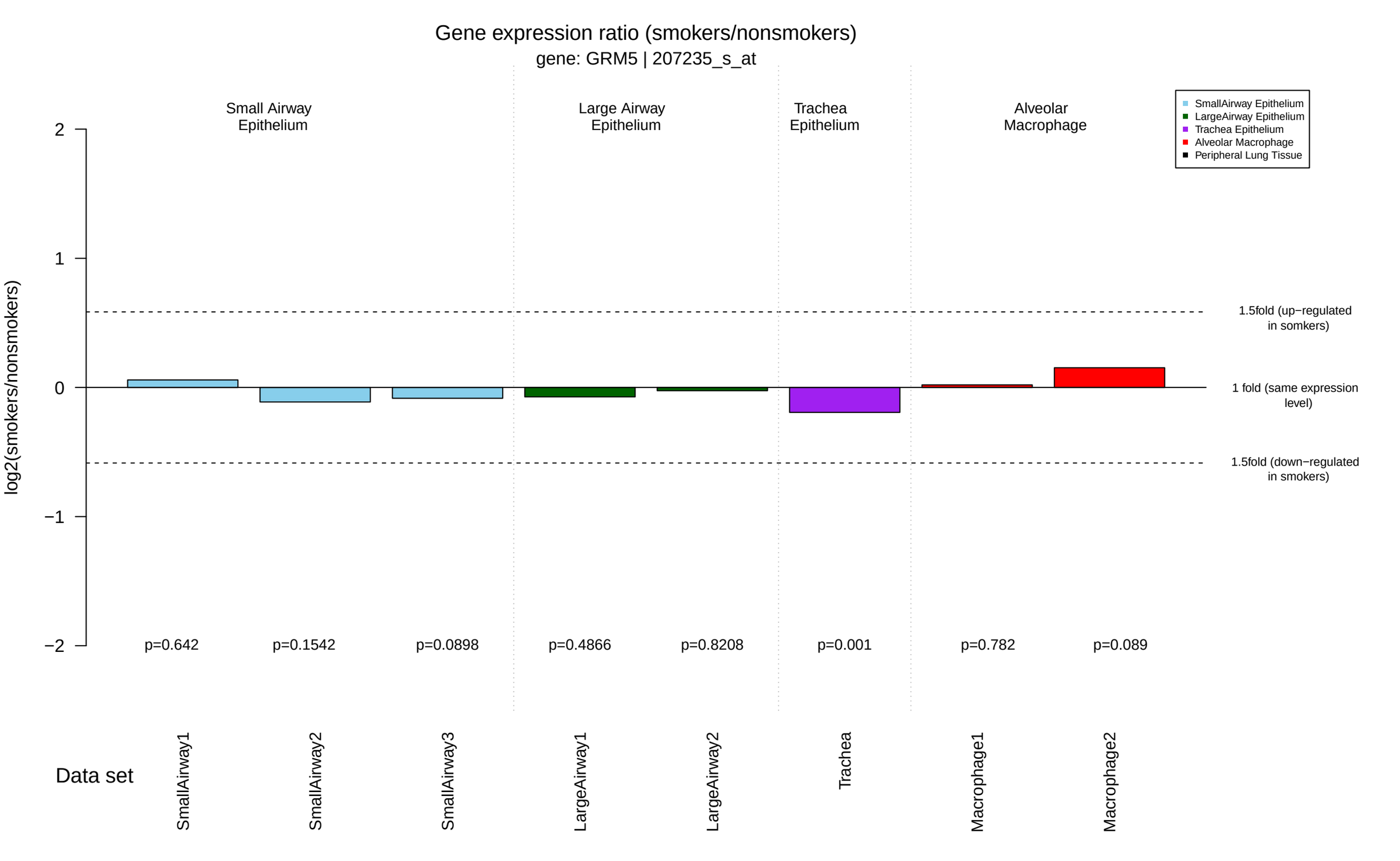
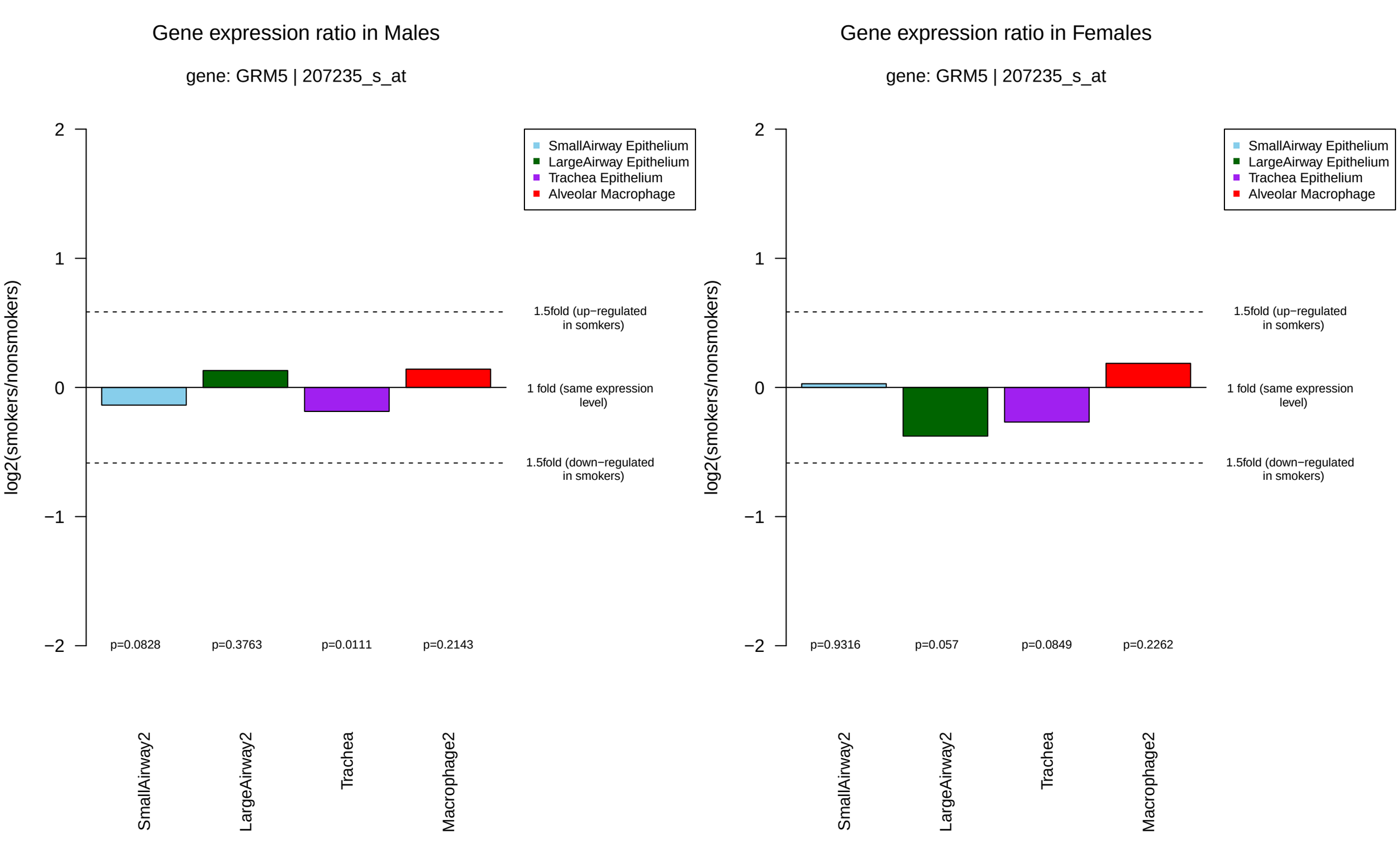
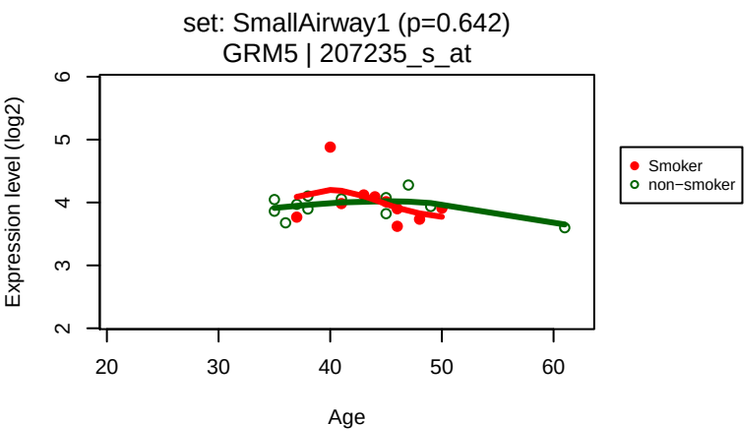
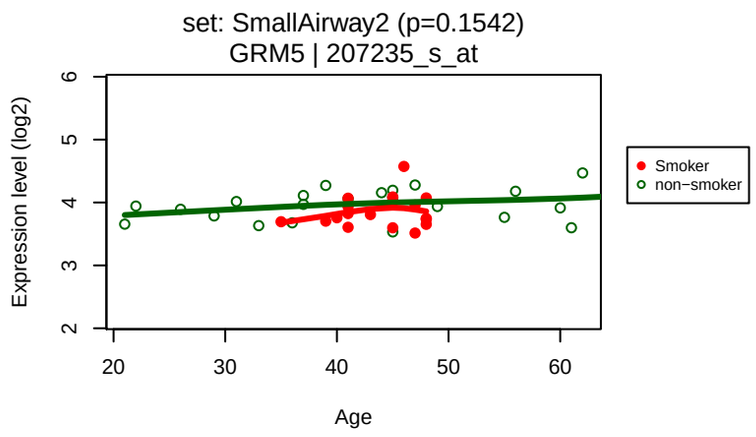
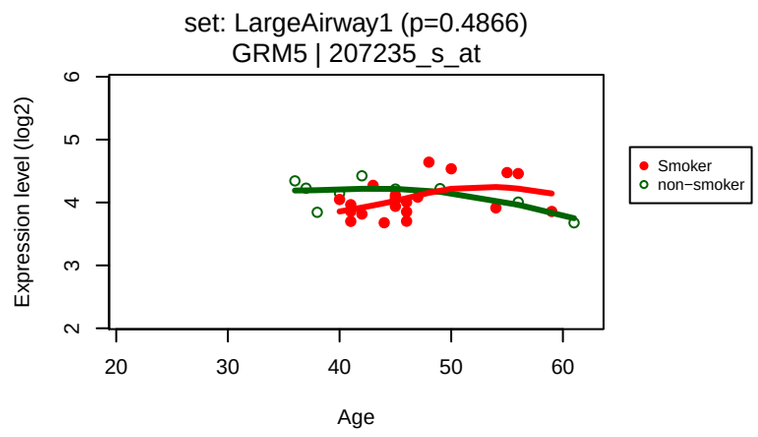

Gene expression of smokers and nonsmokers
gene: GRM5 | 207235_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.642
p=0.1542
p=0.0898
p=0.4866
p=0.8208
p=0.001
p=0.782
p=0.089
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

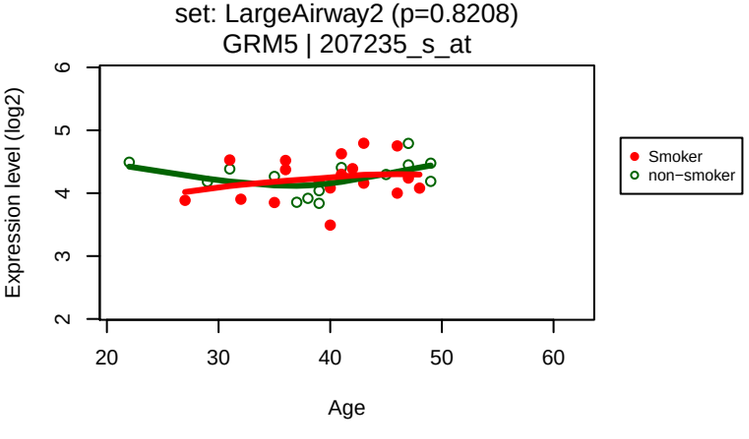

Gene expression ratio (smokers/nonsmokers)
gene: GRM5 | 207235_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.642
p=0.1542
p=0.0898
p=0.4866
p=0.8208
p=0.001
p=0.782
p=0.089
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

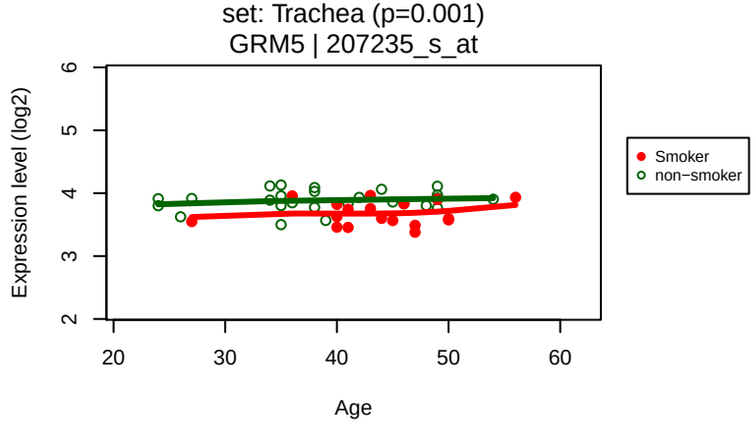

Gene expression ratio in Males
gene: GRM5 | 207235_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0828
p=0.3763
p=0.0111
p=0.2143
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GRM5 | 207235_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9316
p=0.057
p=0.0849
p=0.2262
SmallAirway2
LargeAirway2
Trachea
Macrophage2

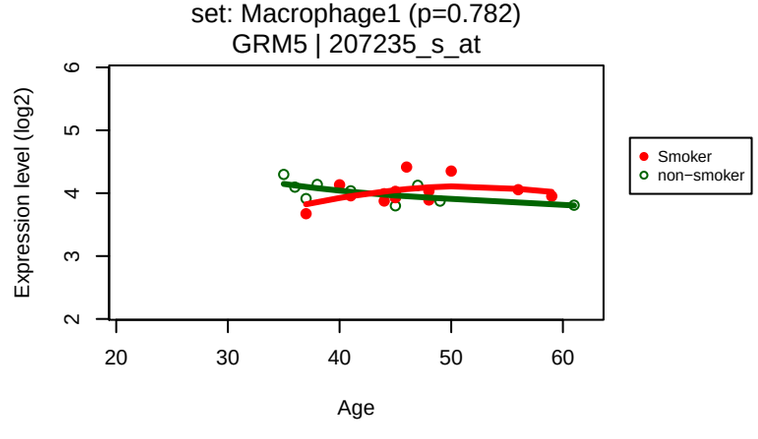

set: SmallAirway1 (p=0.642)
GRM5 | 207235_s_at
Expression level (log2)
Age
Smoker
non−smoker

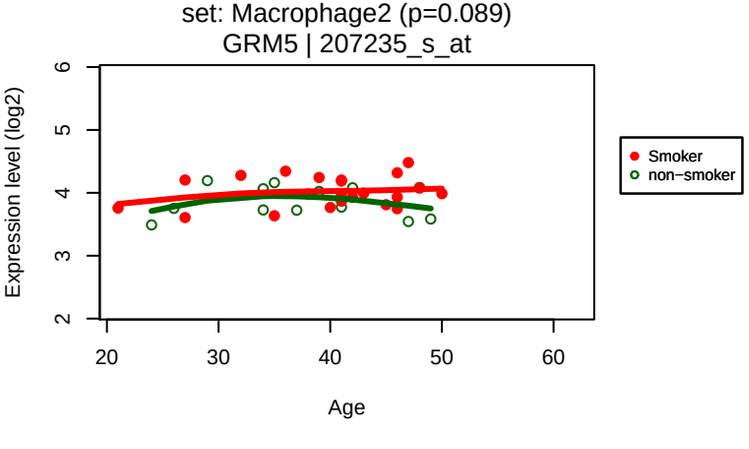

set: SmallAirway2 (p=0.1542)
GRM5 | 207235_s_at
Expression level (log2)
Age
Smoker
non−smoker

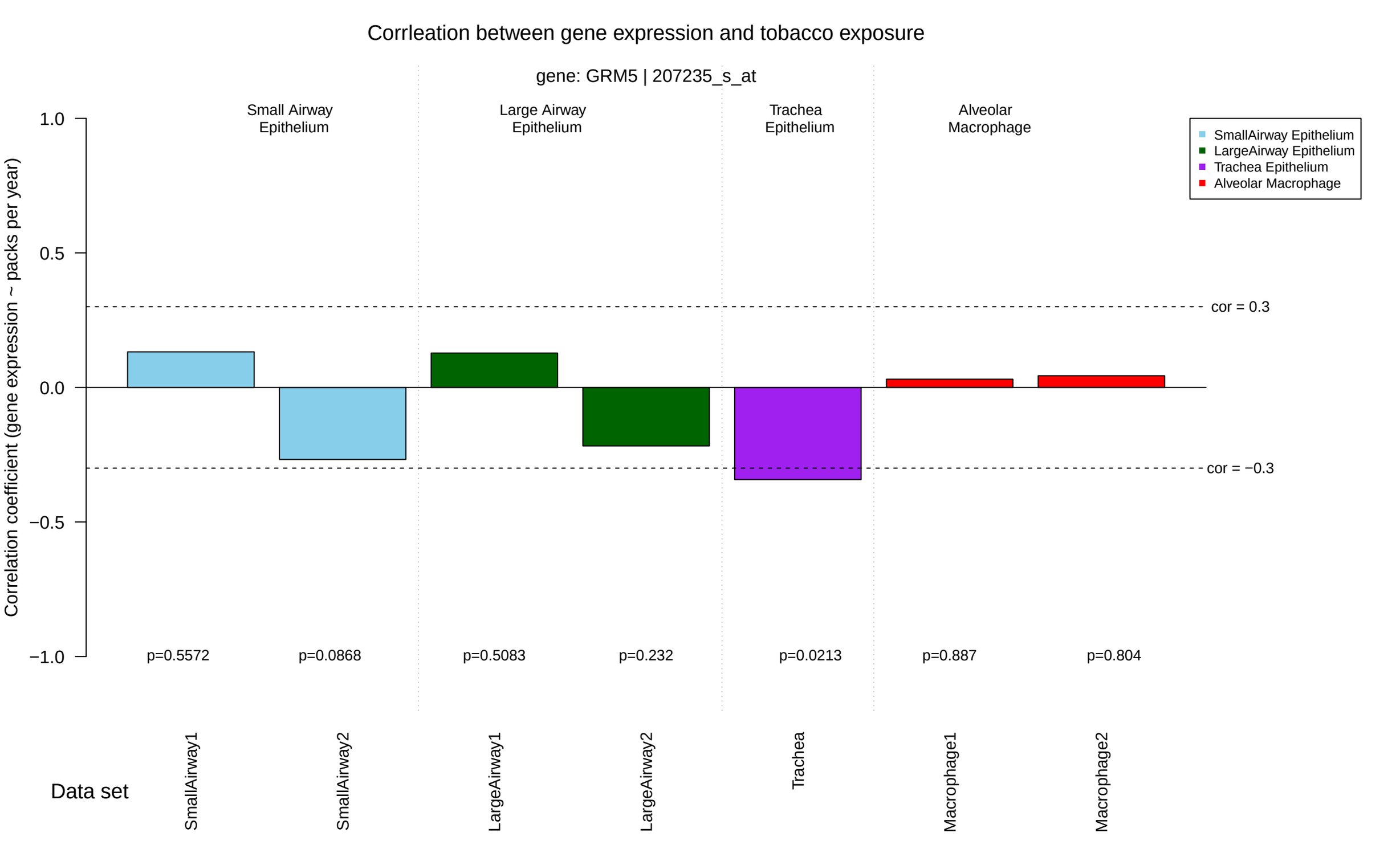

set: LargeAirway1 (p=0.4866)
GRM5 | 207235_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8208)
GRM5 | 207235_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.001)
GRM5 | 207235_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.782)
GRM5 | 207235_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.089)
GRM5 | 207235_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GRM5 | 207235_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5572
p=0.0868
p=0.5083
p=0.232
p=0.0213
p=0.887
p=0.804
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2